Bits of DNA

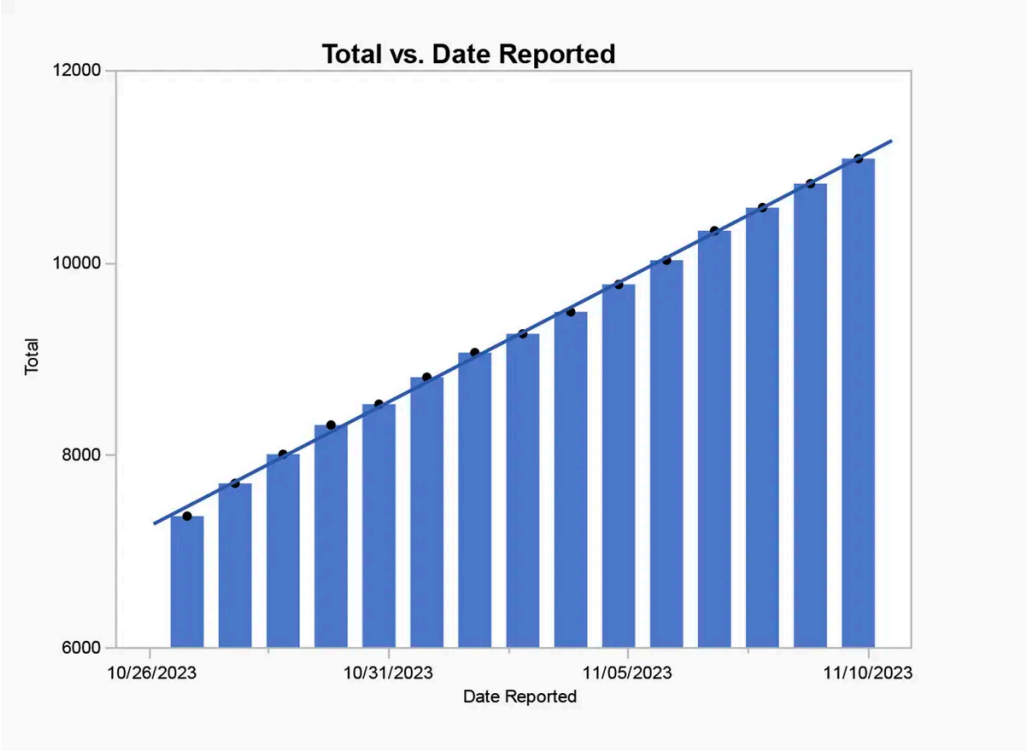
Reviews and commentary on computational biology by Lior Pachter

A note on “How the Gaza Ministry of Health Fakes Casualty Numbers”

March 8, 2024 in reviews, Uncategorized | Tags: Abraham Wyner, Coefficient of determination, linear regression, R2 | by Lior Pachter

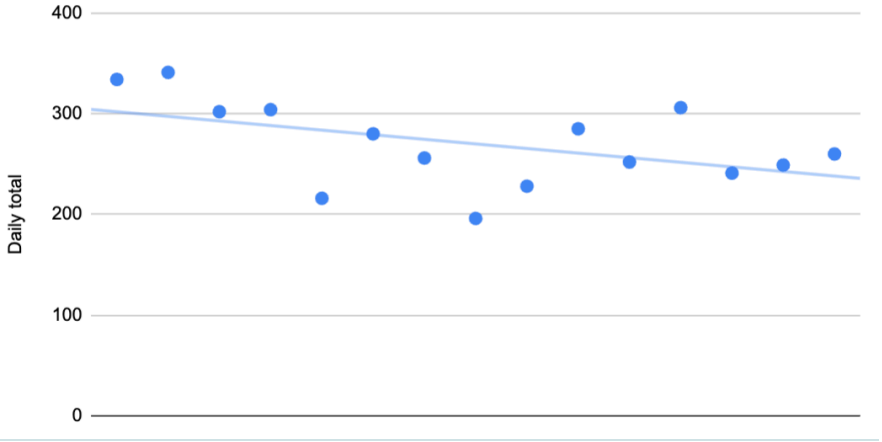
In a Tablet Magazine article titled “How the Gaza Ministry of Health Fakes Casualty Numbers” posted on March 6, 2024, Professor of Statistics and Data Science Abraham Wyner from the Wharton School at the University of Pennsylvania argues that statistical analysis of the casualty numbers reported by the Gaza Ministry of Health is “highly suggestive that a process unconnected or loosely connected to reality was used to report the numbers”.

In the post, he shows the following plot



which he describes as revealing “an extremely regular increase in casualties over the period” and from which he concludes that “this regularity is almost surely not real.”

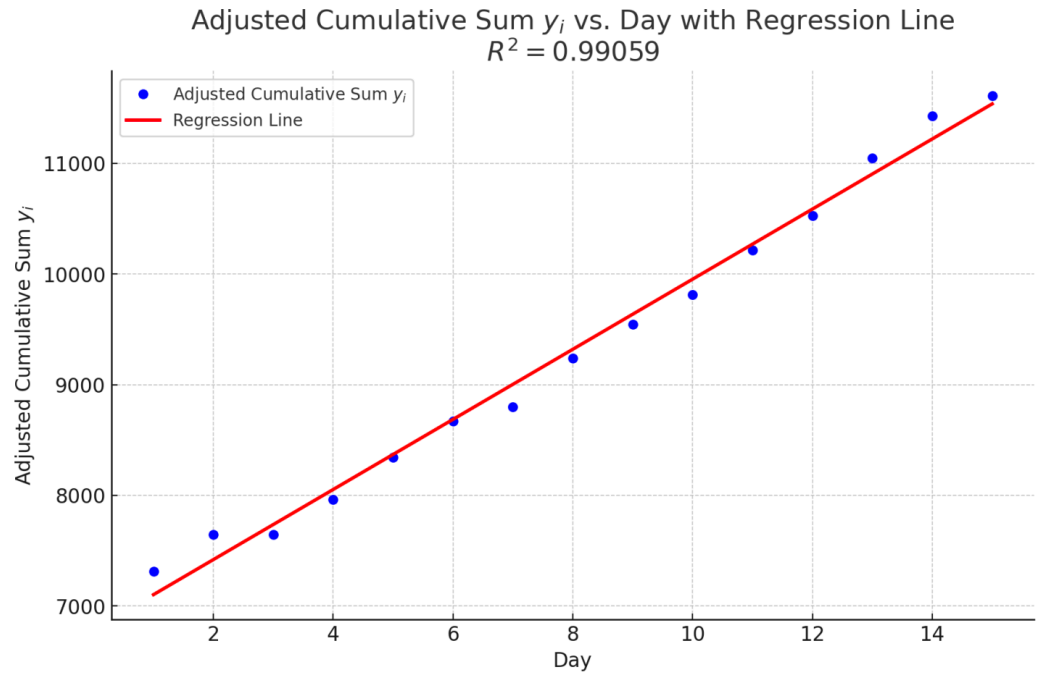
Wyner’s plot shows cumulative reported deaths over a period of 15 days from October 26, 2023 to November 10, 2023. The individual reported deaths per day are plotted below. These numbers have a mean of 270 and a standard deviation of 42.25:



The coefficient of determination for the points in this plot is $R^2 = 0.233$. However, the coefficient of determination for the points shown in Wyner’s plot is $R^2 = 0.999$. Why does the same data look “extremely regular” one way, and much less regular another way?

If we denote the deaths per day by $x_1, \ldots, x_{15}$, then the plot Wyner shows is of the cumulative deaths $y_1 = x_1, y_2 = x_1 + x_2, y_3 = x_1 + x_2 + x_3, \ldots, y_n = x_1 + \cdots + x_n$. The coefficient of determination R^2, which is the proportion of variation in the dependent variable (reported deaths) predictable from the independent variable (day), is formally defined as $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$ where SS_{res} is the sum of squares of the residuals and and SS_{tot} is the variance of the dependent variable. Intuitively, R^2 is a numerical proxy for what one perceives as “regular increase”.

In the plots above, the SS_{res} are roughly the same, however SS_{tot} is much, *much*, higher for the y_i in comparison to the x_i. This is always true when transforming data into cumulative sums, and is such a strong effect, that simulating reported deaths with a mean of 270 but increasing the variance ten-fold to 17,850, still yields an “extremely regular increase”, with $R^2 = 0.99$:



41 comments

Comments feed for this article

March 9, 2024 at 11:32 am
huntercutting

Thank you. You obviously put a LOT of work into your blog offerings. Please know it is appreciated here.

Reply

March 9, 2024 at 12:31 pm
Truth

Thank you, Lior. I have not always agreed with you, but I have great admiration for your rigor and moral clarity. Thank you for not taking the cowardly route that I and most others take.

Reply

March 9, 2024 at 9:58 pm
Luis

🤗 and it means?

Reply

March 10, 2024 at 8:40 am
John

Lior, thank you for your insights. What is your interpretation of the variability between women/children casualties and lack of variability between men/women casualties that he writes about later in the article?

Reply

March 10, 2024 at 10:51 am
Lior Pachter



I don’t know. There could be many reasons for these correlations. Maybe it’s an artifact of the age threshold for children and the distribution of age in Gaza. Maybe it’s the result of lags in recording deaths. Maybe it’s a happenstance arising from so few datapoints. Maybe the data was indeed faked.

I’ll note that there are all sorts of anomalies one can grasp onto. I noticed, for example, that the average, 270 is an integer. Adding up 15 random numbers and then dividing by 15 is unlikely to yield an integer. But it can happen (7% of the time). When one starts floating tons of hypotheses, especially with little data, evidence for one of them doesn’t carry significance.

Reply

March 12, 2024 at 6:55 am
RC

Lior the point is not the abundance of anomalies but their inherent contradictions. No correlation between child and women deaths, negative correlation between men and women deaths and daily variance of total death of 15%. What natural process would generate

these statistics? Are women targeted specifically by the IDF? Is every man surrounded by multiple women? Are women not found together with children in a war zone? When less men die and more women die, shouldn’t the number of child death increase as well? These stats are contradictory. Thats the point of the article, not any one itself alone.

You point at potential “age cuts” but don’t explain how that explains what we see here.

A poor choice of the accumulated graph doesn’t prove or disprove anything other than the author or editor playing the PR game. He did write explicitly that the variance was 15% and thus he was not misleading.

I left out terms such as Statistical Significance, I assume we all understand what that means and where it applies.

March 10, 2024 at 8:44 pm

Ken M

If you look at the numbers, it’s very clear that they update fatalities faster than the update #women or #children (and they don’t specify #men, that is just (#fatalities – #women-#children)). On some days fatalities update but there is no change in the #w or #c; on other days the increase in (#w+#c) exceeds the increase in #f. In other words, in the conditions of war, it is hard to get information. The Gazan Ministry of Health (GMH) makes a list of the name and ID # of every identifiable death; Israel maintains the registry of ID #’s so GMH can’t fake that. That’s why their numbers come out accurate. But in real time, they may get a number of fatalities from a hospital and get the names, which allow identification of #w or #c, only later, maybe much later. And if they get the list of names, they have to go through the registry to determine who is a child or an adult, and maybe for ambiguous names who is a woman or a man, and that probably takes time too. So #w and #c get updated with arbitrary lags, sometimes multiple days worth may suddenly get updated at once. So looking at day-by-day movements of these #’s is meaningless.

I’ll add two other things. First, he says there is no correlation between increment in #women and increment in #children, just like Lior showed that there is no correlation between increment in #fatalities and time. But if you look at the cumulative #women vs the cumulative #children, you get perfect correlation, R^2=0.99 (I checked), just like he finds perfect correlation between cumulative #fatalities and time. Second, for his day-by-day anticorrelation between women and men: because they don’t specify men, only #w and #c, and because they may update in bunches, when there is an update of a lot of women, it will look like there’s not many men (i.e. change in fatalities – change in (women + children) is small, or even negative). When there’s an update where they don’t know the identities so it looks like there’s

no increase in the #women, it will look like there’s a big increase in men – all the fatalities will appear to be men. So that’s why you get an anticorrelation between #women & #men.

Reply

March 11, 2024 at 1:03 am
Ken M

You can see the counts at https://www.ochaopt.org/content/hostilities-gaza-strip-and-israel-reported-impact-day-21

That’s far 21, the 1st day used in the Atlantic article. Vary the last number in the URL from 21 to 35 and you’ll find all the data he used in that article (there’s also a link at the end of the article), but the data keeps going, now above day 140 (with some days with no report).

Reply

March 14, 2024 at 7:51 pm
Lisa Williams

You are exactly right. Here is an independent compilation of data from Gaza authorities.
https://data.techforpalestine.org/docs/casualties-daily/

Usually the number of total deaths are the same, but often the number of women and children are different, because they probably pulled the data at a different time of the day than UN OCHA. So total, women, and children are being updated at different times. Wyner assumed the number of men from the total minus women and men. A daily comparison of total, sex, and age should have never been made with these data.

Reply

March 21, 2024 at 11:06 am
Wes J

Lisa you’re spot on, good eye. I’ve been gathering the TFP numbers for https://data.techforpalestine.org and the women & children are reported by the Gaza Media Office alongside other non-health statistics at different times from the Gaza Health Ministry, which reports daily the total number killed and injured without breakdowns. Lately they’ve been coming in at a fairly regular cadence but no more than once per week. To use them to derive daily totals by gender or age from the total is questionable without having a better idea of the reporting periods they’re using, as you’ve indicated. It’s unclear to me how much they overlap.

Worth noting that the Gaza Health Ministry’s reporting framework in the North ceased entirely in November which is in the middle of the timeframe Wyner chose to focus in on when conducting his analysis. I imagine collecting and reporting on human suffering at this scale is hard enough at any time, but in the chaotic first couple months I can tell from the reports I’ve

gone through that the officials there were still figuring out how to report on this and how often.

March 12, 2024 at 9:27 am
Mehmet Seflek

> The analysis by Abraham is so flawed I’m having difficulty accepting that it is not deliberately disingenuous.

Take basically any distribution of non-negative integers, sample from it, and then take a cumulative sum, and you’ll get an R^2 >0.99. Here’s a simple python script that’ll do it for a Poisson distribution, but you can swap in a different distribution to check: https://pastecode.io/s/e8i2p0z7

you could even prove it mathematically, but that exercise is left to the reader 🙂

Reply

March 12, 2024 at 10:37 pm
A. Berger

> Actually, your critic of Wyner is so flawed I’m having difficulty accepting that it is not deliberately disingenuous.

He is never using R^2 for his argument. His sole numerical argument (concerning the numbers discussed herein) is based on the low covariance of the daily reported deaths. The cumulated figures are used for graphical display only.

Please read carefully this citation of Wyner: “This regularity is almost surely not real. One would expect quite a bit of variation day to day. In fact, the daily reported casualty count over this period averages 270 plus or minus about 15%. This is strikingly little variation. There should be days with twice the average or more and others with half or less.”

His argument is _never_ about R^2 of the cumulated numbers. His argument is about the low covariance of the un-cumulated numbers, which is only 15% of it’s mean.

Show me a Python script that proves “that basically any distribution of non-negative integers” shows a covariance of 15% of it’s mean (or less) and I will sincerely apologize to you. Otherwise you should apologize to Wyner for your gross misinterpretation.

Reply

March 13, 2024 at 9:59 am
Mehmet Seflek

> Show me a Python script that proves “that basically any distribution of non-negative

> integers” shows a covariance of 15% of it’s mean (or less) and I will sincerely apologize to you. Otherwise you should apologize to Wyner for your gross misinterpretation.

Well, you could just input 270 for the poisson parameter in the script I shared. For a Poisson distribution that would be both the mean and the variance, and it’ll give you a mean of 270 and a standard deviation of 16, which is within 5% of the mean, less than the 15% benchmark that you and Abraham Wyner set. David below also pointed this out.

The general point is you could always cherry pick data to support a particular hypothesis. I would hope that a stats academic would hold themselves to a much higher bar before making statements like “The numbers are not real.” There’s more data as David posted below, and there are many other well-known tests you could run to test the null that the data is “naturally” distributed, rather than just saying “the variance is less than I expect due to some heuristic I used.”

March 12, 2024 at 10:44 pm
A. Berger

Erratum: Please replace “covariance” with “standard deviation” in my previous comment.

Reply

March 12, 2024 at 10:59 pm
Lior Pachter



Actually, Wyner is very clear in explaining how he arrives at his conclusion. The following is a quote from his piece:

“Taken together, what does this all imply? While the evidence is not dispositive, it is highly suggestive that a process unconnected or loosely connected to reality was used to report the numbers. Most likely, the Hamas ministry settled on a daily total arbitrarily. **We know this because the daily totals increase too consistently to be real**.” (emphasis mine).

March 12, 2024 at 11:35 pm
A. Berger

Answer to Lior Pachter (for some reason I cannot answer directly):

Yes, the emphasized citation is correct. And how does Wyner substantiate his verbal claim “daily increases are too consistent” numerically? He does so by pointing to the low variance of these very “daily increases”. He does not use the cumulated numbers, and he does not use R^2.

So all the mockery about his allegedly flawed application of R^2 on cumulated numbers is in vain (at best) or deliberately misleading (at worst). Why don’t you comment on the low variance of the daily increases instead?

March 16, 2024 at 12:24 am
Izzy

I mean this as a comment to Mehmet Seflek. You keep pointing to a Poisson distribution, but I think the implicit point of Abraham Wyner is that the distribution should be a complex sum of (dependent) random variables because military operations happen sporadically. In fact, the distribution following a Poisson distribution is exactly what makes the data seem fake. I think it is a reasonable a priori assumption that the distribution should be more complex than that (some days have significantly more deaths than others).

March 12, 2024 at 10:52 am
David McQuillan

There is a discussion on Wikipedia about this

https://en.wikipedia.org/wiki/Wikipedia:Reliable_sources/Noticeboard#Tablet

including a link to dates around those given which show they were cherrypicked

https://docs.google.com/spreadsheets/d/1TNL1kx13vPC6KmpB8CFeEqjjj-ea7DP6X4BFOGsD_ew/edit?
https://docs.google.com/spreadsheets/d/1TNL1kx13vPC6KmpB8CFeEqjjj-ea7DP6X4BFOGsD_ew/edit#gid=0

Reply

March 13, 2024 at 3:33 am
David McQuillan

His argument about low variance in the number of deaths is wrong. If the number of deaths followed a Poisson distribution then the variance would be the same as the mean, and the standard deviation would be the square root of that. That gives a mean 270, variance 270 and std deviation 16.4. The 15% standard deviation is a standard deviation of 270*0.15 = 40.5, actually it is 42 in the data. That is quite a bit higher and really one needs an explanation of the high variance rather than low variance. Which can easily be given by batches people being killed at a time and changes in targets of the bombs.

Ken M’s explanation above of the anti-correlation of deaths between men woen and children seems right to me, batching the checking looks like it takes two days to do.

The only strange thing I can see is that the number of men is lower than expected. One would expect more men killed than women especially as there

would be more of them in essential services – so more would probably be killed even ignoring Hamas militant deaths. If the bombs are not well targeted one would only expect perhaps three percent of the men killed to be a Hamas militant. As to militants I’d expect them to be well protected and so a large number of the dead ones would be under rubble or in tunnels.

The number of men killed in later figures is quite a bit higher than that for women, but the ratio still lower than I would expect in a war.

There’s twice as many children as women in Gaza, but the count of children killed is quite a bit less than twice that for women. I really can’t think why except the missing ones weren’t taken to hospital or just weren’t reported to the authorities.

There is a link in another Wikipedia discussion there that seems to show that even early on the health authority may have been missing a third of the total deaths, see

https://reliefweb.int/report/occupied-palestinian-territory/quality-data-recording-gaza-potentially-evidencing-genocide-frustrated-israeli-strikes-contravening-international-court-justice-orders

UNHCA checked the data released on October 26 with the number dead it knew of within its thousands of staff and families and found only 63% of the deaths had been registered.

Reply

March 14, 2024 at 8:49 am

A. Berger

Thank you and Ken M for your comments, which highlight a key weakness in Wyners analysis. He ignores the fact that the people actually killed are reported with different time delays, depending on many circumstances. And even worse: he does so implicitly without an explanation.

I would like to add another key criticism about his claim that the variation of daily death numbers is too weak. He states that “There should be days with twice the average or more and others with half or less.” But why? He gives no reason for this statement. Again, even worse, he admits that he actually does not know: “Unfortunately, verified control data is not available to formally test this conclusion”. And still, without reliable justification, he claims that “the details of the daily counts render the numbers suspicious.”

Bottom line: There are major flaws in Wyners analysis that should be criticized. It is really a pity that this blog entry of Lior Pachter misses them all. He tries to ridicule the mathematical expertise of Wyner, which backfires.

Reply

March 14, 2024 at 8:55 am
M A/K/M

The rag has cherry-picked data from a certain date-range — in addition. So no need to be shocked that a climate denier would be prone to doing that. In addition to multiple other factors about the genocidally-occupied Gaza Strip which address the concern for anybody with justifiable level of scepticism.(

Which, unless you suspect the other side’s figure, is already on extremely-thin ice.)

Reply

March 16, 2024 at 12:36 am
Izzy

I don’t see why the deaths would follow a Poisson distribution, in fact I think the criticism is that the distribution of deaths should be some complicated mixture (given how military operations timing and tactics are based on complex considerations (including the weather)). Assuming a complex distribution for deaths a priori is reasonable and even laymen find skeptical.

Reply

March 16, 2024 at 6:38 am
Lior Pachter



The variance in this data is 1785. The variance if it was Poisson would be 270.

March 16, 2024 at 11:26 am
Izzy

Hi Loir. It seems odd the data would follow a Poisson distribution when this is a dependent time series. I think that is the crux of the skepticism Dr. Wyner has (and even laymen intuitively have) with the reported data.

March 16, 2024 at 11:54 am
Lior Pachter



The data is not Poisson. If it was the variance would be about 270. But the data is way over dispersed. To the extent that the variance is much higher. In fact more than 6 times higher.

March 16, 2024 at 11:34 am
M A/K/M

@Izzy: You meant like his climate-denial acrobatics? Besides.. It’s our job to take him what he says at his word, not to treat him with kid’s gloves and try to speculate what he actually meant. Unless you meant he’s not supposed to be a professional, just some twisted sort of court-jester.

March 16, 2024 at 12:29 pm
Izzy

Hi Lior. Low variance is one concern (and I see your point that the descriptive

graph he uses masks the degree of variation), but the main concern that deaths do not seem to be following an erratic time series process that one would expect is valid cause for skepticism.

March 16, 2024 at 1:54 pm

Lior Pachter



Again, the baseline model for count data posits a Poisson distribution on the numbers, which in this case would represent a variance of 270. A compound Poisson process makes more sense in this case, and such a process would have even higher variance. Here we see exactly that, a variance of 1785, which is more than 6x what one would see in a Poisson process. If the author thinks the variance should be even higher than *that*, he needs to provide an argument for why, and point to historical data. Of course in this case the variance is even higher, because he appears to have “cherry” picked the 15 days.

But in any case, certainly the counts are not “extremely regular” as he has claimed.

March 16, 2024 at 3:23 pm

David McQuillan

Using a compound Poisson distribution could explain why the variance is so much higher than a Poisson distribution given that it looks like the bombing was quite regular in that period. To do that I think one would have to have something like if a bomb killed anybody then it killed 6 people on average. So of the perhaps 15,000 bombs dropped in the first month killing about 7,500 people – rounding for simplicity – on average eleven out of twelve bombs killed nobody and the twelfth killed six people. I guess that is about possible or somewhere near the truth with people forming groups. Anyway the bit in the article about it being too regular is nonsense. I wonder now if they’ll go on about it being too irregular!

March 16, 2024 at 2:35 pm

Izzy

Again, the argument that the process should follow a Poisson distribution (or sum of Poisson random variables) seems odd. This is a time series random variable that could be modeled with an autoregressive moving average plus other time series variables and their lags (so it’s value over time should not be so predictable). It is reasonable to assume the deaths over time to look more spermatic then it does and I think even laymen come to this conclusion (although you could be right and the 15 days chosen could be an exception). If you wanted to convince me this is cherry picking, does a different time period show more spermatic death counts?

October 19, 2024 at 8:26 pm

Michael Rorer



“…given how military operations timing and tactics are based on complex

considerations (including the weather)” -typically but not in this war. According to the Chief of IAF Tishler interview aand israeli military officers interview in the beginni9ng IDF was dropping thousands of bombs “day and night” erazing entire neighborhoods. Specifically “hunting” militants in their homes. This bombing campaign providing surprising consistency in # of killed and the high women/c – men ratio.

October 19, 2024 at 8:16 pm
Michael Rorer



one small point on the relatively small #of men. There was an interview conducted with several members if Israeli intelligence who worked with target-defining AI system Lavander. here are excerpts from it “… **“we were not interested in killing [Hamas] operatives only when they were in a military building or engaged in a military activity “on the contrary, the IDF bombed them in homes without hesitation, as a first option. it’s much easier to bomb a family’s home. the system is built to look for them in these situations.”** – thid seems to almost guarantee a high proportion of non-combatants. There are other statements they made that point to the same result.

Reply

March 15, 2024 at 1:02 am
David McQuillan

This rather grim business called ‘cemeteries of numbers’ may a reason why the number of male deaths is not as high as one might expect.

https://jacobin.com/2023/11/israel-palestine-gaza-corpse-politics-human-rights-mourning#:~:text=They%20were%20buried%20in%20what,file%20for%20ea

Reply

March 15, 2024 at 6:59 pm
J

Interesting enough, looked into a little bit of Dr. Wyner’s research and his research is focused on statistical analysis using limited data.

Reply

March 16, 2024 at 5:48 am
David McQuillan

Yes he is obviously good at it – and he’s produced this trash. It must be deliberate. I feel dirtied and annoyed by it.

Reply

March 16, 2024 at 8:21 am
J

Or maybe they are both correct within their respective field of expertise. Dr.

Pachter deals with very large datasets, so it makes sense for him to approach it this way. Dr. Wyner deals with limited datasets, and makes risky decision based on these datasets (and it looks like with accuracy, considering he is a consultant to both law, investment banking, and other financial institutions). Are there not many different statistical models to account for different properties?

March 16, 2024 at 11:17 am
David McQuillan

If he would like to actually explain how the variance should actually be much greater than it actually is in the range he chose, never mind if you expand that range by just one day either way, rather than basically just saying look at that, then I will certainly stand in awe of his professorial acumen. Till then I’ll stick with the fairly basic statistics above and I’m fairly sure Lior Pachter will remain satisfied his field of expertise is able to cope.

Reply

March 18, 2024 at 5:21 am
lewi

There could be good reasons to choose this particular time period. First, according to Epstein’s tables, most of the OCHA-UN data is only available for children, women and men in this time period. (Wyner mentions he got the data from Aizenberg, who from what I can understand only used the OCHA-UN data.) The exceptions turn out to be short periods with what seems to be nonstationary or weird behaviour. A few days data exists before October 26, but the ground forces started up operations on October 26 and this could make a good start time. Lewi

Reply

March 18, 2024 at 8:02 am
David McQuillan

He said ‘This regularity is almost surely not real. One would expect quite a bit of variation day to day.’ He chose the time to be regular. And even so he actual variation was far more than one could expect from pure chance if the deaths were independent and random – never mind his ‘This is strikingly little variation.’ If anything an explanation was needed for the higher variation. Lior Pachter gave a possible one above and I expanded on it. The analysis is pretty simple but Professor Abraham Wyner did not do it, he just said look at it, look at it. And as for the negative correlations the business about checking the numbers for the different groups explains that perfectly – they only had four people on the job gathering the statistics from the hospitals and red cross and checking them on spreadsheets. That they batched the different groups and took about two days for that sort of thing is a perfectly reasonable explanation.. The man is supposed to be a professor of statistics, this sort of thing should be pretty obvious to him. This was a hatchet job by a professional expert witness trying to make a case for

his side. You can see what Professor Michael Mann says of him at https://twitter.com/MichaelEMann/status/1766895850670751839.

Reply

March 26, 2024 at 4:26 pm
flyingmonkey



https://factifada.nigelparry.org/wp-content/uploads/2023/12/image-66.png

https://factifada.nigelparry.org/wp-content/uploads/2023/12/image-19-1024×587.png

Here are two graphs of the (almost) first three months of deaths, sources here:

https://factifada.nigelparry.org/resources/casualty-and-other-statistics-from-israels-war-on-gaza/

All that Wyner has done, and the politicized tone and language of his article should be a red flag all of its own, is that he has intentionally presented the data in a simplistic way so that it looks artificial to any viewer.

The Gaza casualty stats are sourced from the same place they were when Israel fully occupied the Strip, when there was Palestinian Authority rule, and since there has been HAMAS rule—hospitals and morgues. That doesn’t change depending on who is in charge. Most of the hospitals in Gaza are run by international organizations, and they have their own statistics which are direct from source.

I’d be really interested to see how Auschwitz death data looked using Wyner’s mathematics. My suspicion would be that it would look very similar. It’s a dirty business, genocide denial, but Wyner was ready to do it. That’s why he published in The Tablet and not a statistical journal.

There have been two peer-reviewed articles about the Gaza death toll published in The Lancet. Les Roberts in TIME magazine did an excellent job of summarizing them, in a recent article that mentione Wyner’s hack job, called “The Science is Clear. Over 30,000 people have died in Gaza”: https://time.com/6909636/gaza-death-toll/

Reply

April 27, 2024 at 11:20 pm
Michael Rorer



a point unrelated to statistical analysis but perhaps explaining some of perceived “anomalies” in data – the particular MO of the army in this war, the way bombing was conducted. It may partly (!) explain the steady flow of civilian (specifically women and children) casualties.

From testimonies of Israeli military intelligence officers:

“We were constantly being pressured: ‘Bring us more targets.’ They really shouted at us,” said one intelligence officer “ and AI system (Lavender) provided a steady flow of targets – low-level Hamas militants and people “linked to Hamas. IDF bombed them in homes without hesitation, as a first option. It’s much easier to bomb a family’s home. The system is built to look for them in these situations.” “we usually carried out the attacks with dumb bombs, and that meant literally dropping the whole house on its occupants…”

Reply

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